New Reactions

1. **Hydrogenation of Benzene (Section 15-2)**

$\Delta H^\circ = -49.3$ kcal mol^{-1}
Resonance energy: ≈ -30 kcal mol^{-1}

Electrophilic Aromatic Substitution

2. **Chlorination, Bromination, Nitration, and Sulfonation (Sections 15-9 and 15-10)**

C6H6→X2,FeX3C6H5X+HXx=Cl,Br

$$C_6H_6 \xrightarrow{X_2,FeX_3} C_6H_5X + HX \quad \mathbf{x = Cl, Br}$$

C6H6→HNO3,H2SO4C6H5NO2+H2O

$$C_6H_6 \xrightarrow{HNO_3,H_2SO_4} C_6H_5NO_2 + H_2O$$

$$C_6H_6 \underset{H_2SO_4,\ H_2O,\ \Delta}{\overset{SO_3,\ H_2SO_4}{\rightleftharpoons}} C_6H_5SO_3H \qquad \textbf{Reversible}$$

3. **Benzenesulfonyl Chlorides (Section 15-10)**

C6H5SO3Na+PCl5→C6H5SO2Cl+POCl3+NaCl

$$C_6H_5SO_3Na + PCl_5 \rightarrow C_6H_5SO_2Cl + POCl_3 + NaCl$$

4. **Friedel-Crafts Alkylation (Section 15-11)**

C6H6+RX→AlCl3C6H5R+HX+overalkylated productR+is subject to carbocation rearrangements

$$C_6H_6 + RX \xrightarrow{AlCl_3} C_6H_5R + HX + \text{overalkylated product}$$
$$\mathbf{R^+ \text{ is subject to carbocation rearrangements}}$$

Intramolecular

Alcohols and alkenes as substrates

$$C_6H_6 \quad + \quad R\overset{\displaystyle OH}{\overset{|}{C}}HR' \xrightarrow[-H_2O]{BF_3,\ 60°C} C_6H_5\overset{\displaystyle R'}{\overset{|}{C}}HR$$

$$C_6H_6 \quad + \quad RCH{=}CH_2 \xrightarrow{HF,\ 0°C} C_6H_5\overset{\displaystyle R}{\overset{|}{C}}HCH_3$$

5. Friedel-Crafts Acylation (Section 15-13)

Acyl halides

$$C_6H_6 \quad + \quad R\overset{\displaystyle O}{\overset{\|}{C}}Cl \xrightarrow[\;]{1.\ AlCl_3 \quad 2.\ H_2O} C_6H_5\overset{\displaystyle O}{\overset{\|}{C}}R \quad + \quad HCl$$

Requires at least one full equivalent of Lewis acid

Anhydrides

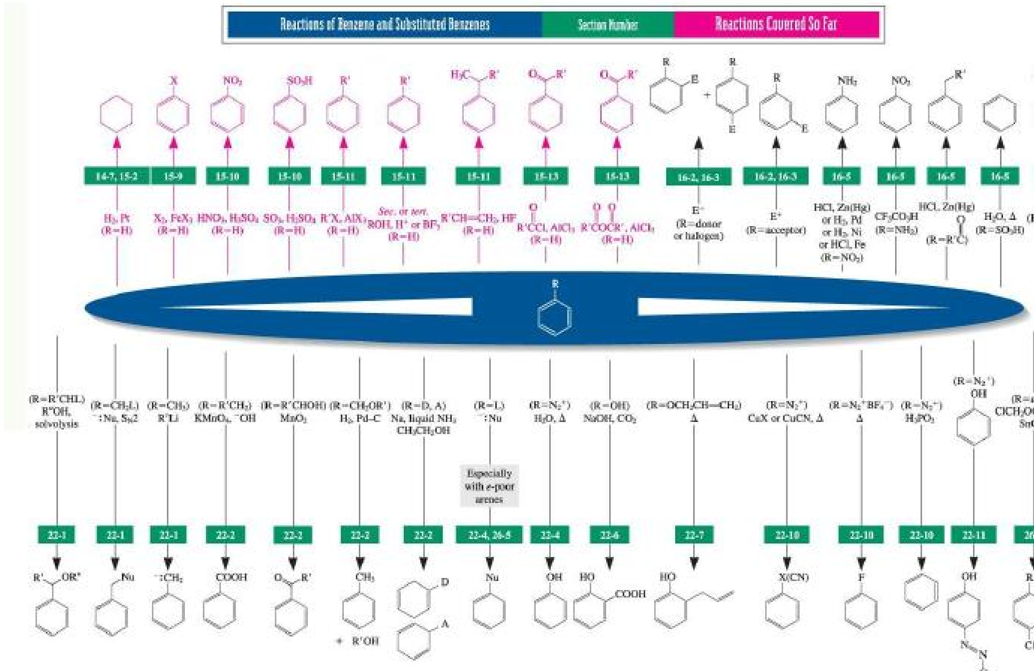
$$C_6H_6 \quad + \quad CH_3\overset{\displaystyle O}{\overset{\|}{C}}O\overset{\displaystyle O}{\overset{\|}{C}}CH_3 \xrightarrow{1.\ AlCl_3 \quad 2.\ H_2O} C_6H_5\overset{\displaystyle O}{\overset{\|}{C}}CH_3 \quad + \quad CH_3COOH$$

Important Concepts

1. Substituted benzenes are named by adding prefixes or suffixes to the stem "benzene." Disubstituted systems are labeled as 1,2-, 1,3-, and 1,4- or **ortho, meta,** and **para,** depending on the location of the substituents. Many benzene derivatives have common names, sometimes used as bases for naming their substituted analogs. As a substituent, an aromatic system is called **aryl;** the parent aryl substituent, C6H5,$\mathrm{C_6H_5}$, is called **phenyl;** its homolog C6H5CH2$\mathrm{C_6H_5CH_2}$ is named **phenylmethyl (benzyl).**
2. Benzene is not a cyclohexatriene but a delocalized cyclic system of six π π electrons. It is a **regular hexagon** of six sp2sp^2-hybridized carbons. All six pp orbitals overlap equally with their neighbors. Its unusually low heat of hydrogenation indicates a **resonance energy** or **aromaticity** of about 30 kcal mol−1 (126 kJ mol−1). $30\,\mathrm{kcal\,mol^{-1}}\ (126\ \mathrm{kJ\,mol^{-1}})$. The stability imparted by aromatic delocalization is also evident in the transition state of some reactions, such as the Diels-Alder cycloaddition and ozonolysis.
3. The special structure of benzene gives rise to unusual UV, IR, and NMR spectral data. 1H NMR$^1\mathrm{H\ NMR}$ spectroscopy is particularly diagnostic because of the unusual **deshielding** of aromatic hydrogens by an **induced ring current.** Moreover, the substitution pattern is revealed by examination of the o, m,o, m, and pp coupling constants.
4. The **polycyclic benzenoid hydrocarbons** are composed of linearly or angularly fused benzene rings. The simplest members of this class of compounds are naphthalene, anthracene, and phenanthrene.
5. In these molecules, benzene rings **share** two (or more) carbon atoms, whose ππ electrons are delocalized over the entire ring system. Accordingly, naphthalene shows some of the properties characteristic of the aromatic ring in benzene: The electronic spectra reveal extended conjugation, 1H NMR$^1\mathrm{H\ NMR}$ exhibits deshielding ring-current effects, and there is little bond alternation.
6. Benzene is the smallest member of the class of aromatic cyclic polyenes

following **Hückel's (4n+2)**$(4n+2)$ **rule.** Most of the 4n$4n$ ππ systems are relatively reactive **anti-** or **nonaromatic** species. Hückel's rule also extends to aromatic charged systems, including the cyclopentadienyl anion, cycloheptatrienyl cation, and cyclooctatetraene dianion.

7. The most important reaction of benzene is **electrophilic aromatic substitution.** The rate-determining step is addition by the electrophile to give a delocalized hexadienyl cation in which the aromatic character of the original benzene ring has been lost. Fast deprotonation restores the aromaticity of the (now substituted) benzene ring. Exothermic substitution is preferred over endothermic addition. The reaction can lead to halo- and nitrobenzenes, benzenesulfonic acids, and alkylated and acylated derivatives. When necessary, Lewis acid (chlorination, bromination, Friedel-Crafts reaction) or mineral acid (nitration, sulfonation) catalysts are applied. These enhance the electrophilic power of the reagents or generate strong, positively charged electrophiles.
8. **Sulfonation** of benzene is a **reversible** process. The sulfonic acid group is removed by heating with dilute aqueous acid.
9. **Benzenesulfonic acids** are precursors of benzenesulfonyl chlorides. The chlorides react with alcohols to form sulfonic esters containing useful **leaving groups** and with amines to give sulfonamides, some of which are medicinally important.
10. In contrast with other electrophilic substitutions, including Friedel-Crafts acylations, **Friedel-Crafts alkylations** activate the aromatic ring to further electrophilic substitution, leading to product mixtures.


Reactions of Benzene and Substituted Benzenes
Section Number
Reactions Covered So Far